Tab 1

[Project title] Design Doc

Status: Draft
Authors: Person , Person
Supplier Company:
Team: *Select 1 GVO team/workflow that you'd like to pilot this with.*
Tracking Buganizer issue: b/
GVO Contributors: Person , Person
Last major revision: Date

1. Overview

Objective

In 1-2 sentences, describe the business goal of this project. At a high level, what should be different once this design (or an alternative) is in place?

What key metrics do you aim to impact? (e.g., AHT, CSAT, Error Rate, Time Saved)

For example: "gShoe should keep the user's feet warm and dry even when they walk through the snow."

This section should not:

- *Describe the problem (use "Background" section below)*
- *Be a detailed list of requirements (use "Requirements" section below)*
- *Propose a solution (use "Design" section below)*

Background

Provide context for an unfamiliar reader to understand the proposal.

- *What was the original Problem Statement?*
- *Why is it important to solve?*
- *What solutions do we already have that solve similar problems?*
- *How does this relate to the GVO innovation theme? (e.g., Do more with less, Drive exceptional user experience, etc)*

This section should not:

- *Be a lengthy explanation with a lot of details. Instead, prefer to link to other docs where available.*
- *Include information about your design*

- *Talk about ideas to solve the problem.*

Scope

IN SCOPE:

- *What is in scope and will be addressed by this solution?*

OUT OF SCOPE:

- *What is not in scope and will NOT be addressed by this solution?*

2. Outcomes & Measurement

Key Metrics:

- *What are the primary metrics that will be used to evaluate the success of this pilot? (e.g., Average Handle Time (AHT), CSAT, Error Rate, Time Saved, Adoption Rate, Revenue Impact, etc.)*
- *For each metric, please specify how it will be measured and the current baseline (if known).*

Target Outcomes:

- *What are the specific target goals for each key metric? (e.g., Reduce AHT by 15%, Improve CSAT by 5 points, Increase adoption to 80% of agents).*
- *What is the timeframe for achieving these targets?*

Measurement Method:

- *How will the data for these metrics be collected? (e.g., Existing dashboards, new tracking within the tool, manual tracking, user surveys).*
- *Are there any dependencies on Google systems or teams to gather this data?*

Qualitative Feedback:

- *Beyond quantitative metrics, how will qualitative feedback be gathered to assess success? (e.g., User interviews, feedback forms).*

Alignment with Objectives:

- *How do these success metrics directly relate back to the project objectives outlined earlier?*

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3. Requirements

Create a prioritized summary of business requirements, **and** provide detailed description for each. If you have multiple use requirements, prioritize them in order of expected impact. Also, indicate any dependencies between requirements.

Priority	User	Requirement	User Story
P0			
P1			
P2			
P3			

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4. Data Access, Security, and Retention

Data Requirements

What specific data, if any, will be collected, processed, or accessed for this project? Please list specific dashboards, tables, data fields, or types of information.

What is the business necessity for accessing/using each type of data?

Does your team currently have access to this data, or will access need to be requested from Google?

Will this solution require access to any Google-internal systems or APIs? If yes, please specify.

Data Use & Minimization:

How will the collected/accessed data be used within the solution?

How will you ensure that only the minimum necessary data is collected, processed, and retained to achieve the project's objective?

Data Storage & Security:

Where will this data be stored? (e.g., vendor's cloud, Google Cloud Platform instance, other). Please describe the storage solution.

What security measures will be in place to protect the data from unauthorized access, use, or disclosure?

Data Access Controls:

Who will have access to this data? Please describe roles or teams.

What controls will be implemented to manage and audit data access?

Data Retention & Deletion:

What is the proposed retention period for this data? Please justify.

What is the process for securely deleting the data when it is no longer needed or at the end of the project/pilot?

Privacy Risks & Mitigations:

What potential privacy risks to individuals or Google have you identified in relation to the data handling in this project? What steps will you take to mitigate these risks?

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5. Dependencies

Discuss your dependencies on other services, data, or tooling. Any reliance on YouTube internal systems, data (including specific tables or APIs), tools (like Cases, YuRT), or team support (e.g., BizTech, ADS). What happens if they're unavailable for a period of time? Distinguish between dependencies that are required for general operation, vs those that provide extra data quality, vs those that are useful for debugging. Which services must be

running for your job to start up? Don't forget subtle dependencies like resolving names using DNS or checking the local time.

Are you introducing any cycles, such as blocking on a service that can't run if your jobs aren't already up? If you have doubts, discuss your use case with the team that runs the service.

Potential Risks

What are the main technical or operational risks you foresee in the development, deployment, or running of this solution?

Are there any dependencies that could pose a risk?

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6. Design

Overview

One page high-level overview; put details in the next section and background in the previous section. Should be understandable by a new engineer not working on the project. Diagrams can be especially useful to quickly convey the shape of the solution. This section does not need to prove that the solution meets the objective or why it is better than alternatives.

Detailed design - Required if shortlisted

You might want to embed code via [code blocks](), like this:

```
None
// Example code can go here.
```

Weigh the pros and cons of the approach you are recommending and provide a justification for why you chose this approach over those discussed in the "Alternatives considered" section below. For smaller details discuss alternatives inline while reserving the "Alternatives considered" section for overall alternative designs.

Make sure readers can find your code. The tracking issue / hotlist listed in the header is fine if CLs are linked to it; if not, list the directories where the code will go.

Alternatives considered

Clearly list the other potential approaches to meeting the objective that you considered and why the current proposal was ultimately selected. In the rare cases where requirements or system constraints only allow for one possible high-level approach, that should be highlighted here and alternatives to specific details should still be discussed in-line in the detailed design section.

An effective strategy is to compare dimensions of the solution space and how they vary across alternatives, such as latency, data staleness, cpu cost, engineering investment, etc. A color-coded table explaining favorability in that dimension and how important it is from green through yellow, orange and red quickly conveys the rationale to a reader, though this is just one possible approach to explain the trade-offs. For example:

	Proposed solution	Alternative 1	Alternative 2
Dimension 1 (e.g. engineering investment)	➕ Degree of favorability (e.g. trivial implementation)	~ Degree of moderate negativity (e.g. 1 SWE quarter)	➖ Degree of negativity (e.g. complete rebuild of new system, 1 SWE year)
Dimension 2 (e.g. resource costs)	~ Degree of slight negativity (e.g. 1K additional GCUs)	➕ Degree of favorability (e.g. 20k GCU savings)	➕ Degree of favorability (e.g. 20k GCU savings)
Dimension 3 (e.g. end-user perceived latency)	~ Degree of moderate negativity (e.g. additional 100ms latency at 99%ile)	➖ Degree of negativity (e.g. additional 5 second latency at 50%ile)	➖ Degree of negativity (e.g. additional 5 second latency at 50%ile)

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7. Solution Risks & Mitigation

Beyond the data considerations outlined in "Data Access, Security, and Retention," please briefly address:

Reliability & Stability: What steps will be taken to ensure the solution is stable and reliable during the pilot?

Scalability Considerations: While this is a pilot, what aspects of the design support potential future scaling if the pilot is successful?

Other Potential Risks: What are the main technical or operational risks (e.g., integration challenges not related to data, platform limitations, user adoption) that could impact the pilot?

Accessibility: How will you ensure the solution is accessible to users, including those with disabilities?

Other Compliance: Are there any other notable compliance or regulatory standards (e.g., industry-specific regulations) being addressed?

Risk Description	***Likelihood (Low/Med/High)***	***Impact (Low/Med/High)***	***Mitigation Plan***	***Proposed Owner (Vendor/Google)***
e.g., Integration issues with [Specific Tool]	*Medium*	*High*	*Conduct early testing with mock APIs.*	*Vendor*

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8. Project management

Timeline

Please provide an estimated timeline for the key phases of the project (e.g., Design, Development, Testing, Pilot).

What are the major milestones and their expected completion dates?

Key Deliverables

What are the main deliverables for this project? (e.g., Working prototype, Final report, Training materials, Technical documentation).